Week 1 Assignment

Introduction to Wireless Communication

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Code Summary:

This project simulates a 6G **Non-Terrestrial Network (NTN)** scenario where a high-altitude transmitter, such as a drone, communicates with a mobile user on the ground. Because the wireless signal must travel vast distances of up to 3 kilometers through the air, it experiences extreme path loss and fading. To overcome this, the simulation integrates a **Reconfigurable Intelligent Surface (RIS)** on the ground. This RIS acts as a smart, programmable mirror that can reflect and boost the drone's signal directly toward the user, helping to extend coverage and maintain connection stability even at the extreme edges of the service area.

To manage this complex environment, the project implements a **Deep Reinforcement Learning (DRL)** agent using a Deep Q-Network (DQN). As the ground user moves, the AI continuously monitors the changing channel conditions and learns to make real-time, joint decisions on the optimal transmit power, data modulation scheme, and RIS phase mode. As proven by the training learning curve, the agent successfully figures out how to dynamically adapt to severe signal drops, finding the best possible balance between maximizing the Signal-to-Noise Ratio (SNR) and minimizing unnecessary power consumption.